

Adapting CHMTRN (CHeMistry TRaNslator) for a New Use

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We have adopted and extended the CHMTRN language and used it for the knowledge base of a computer program to generate a large database of synthetically accessible, drug-like chemical structures, the Synthetically Accessible Virtual Inventory (SAVI) Database. CHMTRN is a powerful language originally developed in the LHASA (Logic and Heuristics Applied to Synthetic Analysis) project at Harvard University and used together with the chemical pattern description language, PATRAN, to describe chemical retro-reactions. The languages have

proved to be useful beyond the design of retrosynthetic routes and have a potential for much wider use in chemistry, and this paper describes CHMTRN and PATRAN as now re-implemented for the forward-synthetic SAVI project, but able to describe both forward and retro-reactions.

INTRODUCTION

Virtual screening has become a common step in early stage drug discovery. However, screenable databases of compounds are still dwarfed by the size of the universe of drug-like molecules.¹ We and others² aim to generate databases of billions of compounds that can be synthesized in one step from readily available starting materials, which constitutes a significant challenge. For our project, the Synthetically Accessible Virtual Inventory (SAVI) Database, it was necessary to create and maintain a generic chemical reactions knowledge base.³ Use of a text-based language was preferred to a graphical tool both because of the potential savings in programming effort and because it would allow anyone to work on, and read, knowledge base content using only a text editor. For a while in the 1970s the use of natural language like programming languages was favoured. The idea went out of fashion but, with the rapid development of advanced natural language processing, it is perhaps time to look at it again.

CHMTRN, originally developed by a team at Harvard University for the LHASA (Logic and Heuristics Applied to Synthetic Analysis) project in the 1970s,^{4,5} has been adopted and adapted for the SAVI project. The purpose of this paper is to describe those parts of the revised language that we are using.

BACKGROUND

When CHMTRN was created by the team at Harvard University it was extremely forward looking. Its coverage and flexibility still do not appear to be matched by any other language for coding reaction and retro-reaction descriptions.

In the LHASA project, a software interpreter packed the information contained in the knowledge base into a set of computer words for fast processing at run time. We have developed a new, similar software interpreter, which we will describe in a separate paper about the SAVI project, using the CACTVS toolkit.⁶ This paper is about the language itself. We have adopted those terms used in the original CHMTRN that are relevant to our project and added new ones, including modifications to make the language more convenient for use with applications carrying out reaction prediction as well as those carrying out retrosynthetic analysis. The definitions of many terms in the language as implemented for SAVI are closely similar to those used in the original CHMTRN, since the terms are everyday ones in chemistry, such as “acid halide”, “double bond”, “leaving group”. Nevertheless, in this paper we have defined how all of them are used in SAVI to avoid potential uncertainty.

CHMTRN AND PATRAN

Each description of a retrosynthetic reaction in CHMTRN is termed a “transform”. A central purpose of CHMTRN is to allow reasoning about sets (e.g. a conditional statement based on whether an atom belongs to a set with a particular property). It provides most of the control and conditional functionality expected of a modern programming language, such as “if ... then .. else”, and “for each” statements. The writer can instruct the application reading the knowledge base to adjust its confidence in the likely success of a reaction, depending on chemical structure and functionality in a query, and to issue messages to the end user.

Transforms in the CHMTRN developed for the LHASA project could contain a linear string description of the keying features of the reaction product (i.e. the target in a retro-reaction) written in PATRAN (*vide infra*).⁷ We have extended PATRAN to describe also the precursors to the retro-reaction, thus allowing transforms to be run in either direction. This was true for LHASA transforms that included “2D” patterns, but the 2D patterns were more difficult to design and enter and provided less support for specifying atom and bond properties than the so-called “1D” PATRAN strings.

LHASA transforms were written retrosynthetically. Transforms for the SAVI project could have been written in the synthetic direction with only the minor change of introducing a forward reaction arrow into the PATRAN language and small changes to how multiple reaction components are handled but we have written them retrosynthetically for backward compatibility with the LHASA program.

Transform, Atom, and Bond Identifiers

The first line of each transform description is a header defining a unique, integer identifier for the transform.

e.g. TRANSFORM 1234

Atoms and bonds that are included in the keying substructure(s) are uniquely identified within a transform by the terms ATOM*n and BOND*n, where n is an integer between 1 and 32. Note that while an atom or bond is unique within a transform, it may be present in both the target (product) and a precursor (reactant). For example the oxygen atoms in the target and the first precursor in the following retro-reaction are the same atom:



Hydrogen atoms

Hydrogen atoms are not normally treated as atoms in their own right in CHMTRN but as a property of the atom to which they are attached. Thus methane is not perceived as four hydrogen atoms each attached by a single bond to the same carbon atom, but as a carbon atom with a hydrogen count of 4. The transform writer can, however, include explicit hydrogen atoms by using the term HYDROGEN*ATOM(S).

The distinction between explicit and implicit hydrogen atoms has implications for how CHMTRN statements should be phrased. Consider the substructure in Figure 1(a), in which the atoms are numbered as shown, the two R groups are non-hydrogen and there are no additional non-hydrogen substituents on atom 2. It might thus be represented as in Figure 1(b), in which “H2” represents two implicit hydrogen atoms on atom 2. In Figure 1(c) one explicit hydrogen atom has been attached to atom 2 and so, to make up unsatisfied valency, there is one implicit hydrogen. The value returned for “THE NUMBER OF HYDROGEN ATOMS ON ATOM*2” is two in case (b) and case (c). However, the value returned for “IF THERE IS A HYDROGEN*ATOM ALPHA TO ATOM*2 ...” is false in case (b) and true in case (c). By default, explicit and implicit hydrogen atoms are all ignored when a statement does not refer directly to hydrogen: e.g. there are two non-hydrogen atoms alpha to atom 2 (R^1 and R^2) and so the value returned for “THE NUMBER OF ATOMS ALPHA TO ATOM*2” is two in case (c) as well as in case (b). The word EXPLICIT*HYDROGEN can be used in order to include explicit hydrogen atoms: “THE NUMBER OF ATOMS ALPHA TO ATOM*2 INCLUDING EXPLICIT*HYDROGEN” would return the value two for case (b) and the value three for case (c).

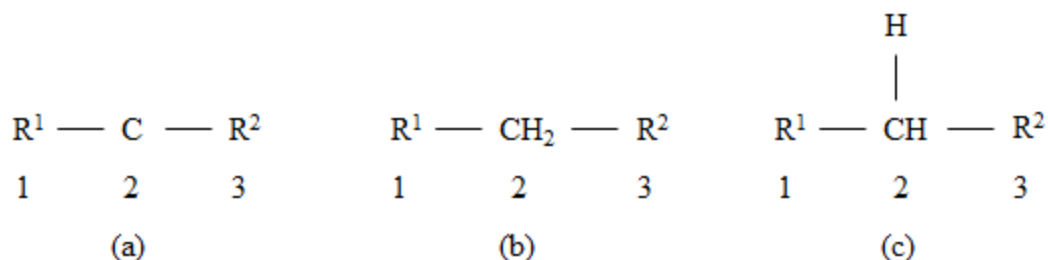


Figure 1. Representation of implicit and explicit hydrogen atoms

PATTERNS

In early implementations of the CHMTRN language, transforms were keyed by functional groups in a target structure (product). Later PATRAN, a flexible, string representation for keying substructures to be read by a PAttern TRANslator module, was developed and incorporated into transform descriptions under the header, 1D*PATTERN. Transforms contained information only about the substructural features associated with the product of a reaction since LHASA operated retrosynthetically. To support operating also in the synthetic direction, a new representation was developed and incorporated into transforms under the header, 2D*PATTERN. As explained below, for our current project we have instead extended the linear 1D*PATTERNS to describe the precursors as well as the products of reactions.

Pattern statements in a transform are preceded by the name of the pattern type and a carriage return. They are followed, after the last pattern, by END*PATTERN or END*PATTERNS. Multiple patterns of the same type are allowed, provided that the atom (and bond) numbering corresponds between them.

1D*PATTERN

The PATRAN 1D patterns used in CHMTRN are similar to other linear representations such as SMILES,^{8,9} SMARTS,¹⁰ and Sybyl Line Notation,^{11,12} and, in fact, PATRAN pre-dates most of them. Branching is indicated using parentheses. Atoms are represented by their elemental symbols in capital letters (e.g. chlorine is “CL”), with the letter X representing any element. Bonds are represented by “-”, “=”, “#”, “%”, “&”, and “+”, for single, double, triple, aromatic, any bond type, and no bond, respectively. Alternative atoms and alternative bonds are separated by commas. Rings are indicated by an @ sign followed by the sequence number of an earlier atom in the pattern string to which a bond connects to complete the ring (atoms and bonds are automatically numbered sequentially from the left).

Atom and bond properties are inserted between square brackets. Alternative values for a property are separated by commas; alternative properties are separated by semicolons (e.g. [HS=0;ARYL=YES]). The atom properties currently supported are listed in Table 1.

For example, the following pattern represents the substructures in Figure 2 (where no substituent is shown in Figure 2, any substituent is allowed).

N[HS=1]%C(-C[FGS=ESTER,AMIDE])%C,N%C%C@1

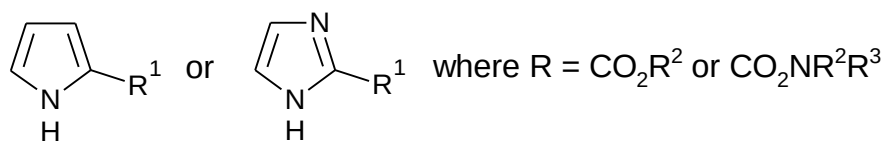


Figure 2. A pair of related generic structures

Long patterns can be written on multiple lines by breaking after a bond symbol, comma, or semicolon.

2D*PATTERN

Many LHASA transforms include code under the header, 2D*PATTERN. When a CHMTRN file is used with LHASA, the purpose of the 2D*PATTERN is to provide keying information about the precursors generated by transforms, in order to support the prediction of reactions (as distinct from retro-reactions). Figure 3 is an example of a 2D pattern from transform 2875 for a Copper[I]-catalyzed azide-alkyne cycloaddition, written recently by Martin Ott.

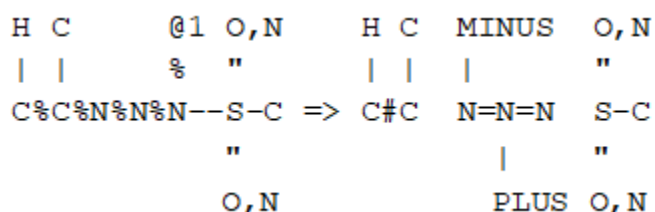


Figure 3. An example of a 2D pattern

2D patterns are difficult to write and maintain, because of the requirement for multiple lines of text to be correctly aligned, and they carry less information about atom and bond properties than 1D patterns. They are included in transforms written for our current project, for backward compatibility with LHASA, but not used by our interpreter, which uses NEW*1D*PATTERNS.

NEW*1D*PATTERN

NEW*1D*PATTERNS convey information about the precursors, as well as the products, of retro-reactions and about atom correspondences between the two sides of the retrosynthesis arrow. The representation of atoms, bonds, and their properties is the same as in 1D patterns that describe only the reaction product (target) but a retrosynthetic arrow is included and precursor patterns are written to the right of it. Within the context only of the pattern description (e.g. for

use with the @ symbol), the atoms and bonds in the set of fragments to the left and right of the retrosynthetic arrow are numbered independently. Atoms to the right of the arrow are mapped to atoms on the left by appending ^ and the number of that atom, counted from the left. It is not necessary to map bonds between the two sides of the arrow since their mapping is implied by that of the atoms. Within CHMTRN statements, atoms in the set of precursors after application of the mechanism retain the numbers they had in the target; if there are new atoms or bonds in the precursors they are given sequence numbers continuing from the last number, respectively, used in the pattern string for the target. Patterns written in multiple lines can be broken following the retro-reaction arrow as well as following a bond symbol, comma, or semicolon.

For example, the following pattern, taken from transform 6026 represents the retro-reaction shown in Figure 4.

S[HS=0](=O)(=O)(-C[ARYL=YES])-N ==>

S^1[HS=0](=O^2)(=O^3)(-C^4[ARYL=YES])-Cl + N^5[HS=1,2]

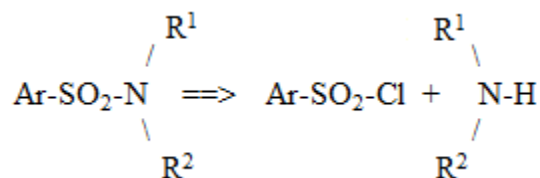


Figure 4. The retro-reaction covered by transform 6026

CHMTRN STATEMENTS

From its inception, CHMTRN was intended to be English-language like and to use chemical terms wherever appropriate.⁴ A set of “buzz words” supports this. For example, the first of the

following statements contains all the key words needed by the interpreter software but the second, making use of buzz words, is clearer to a human:

```
OXIME SUBTRACT 30
```

```
IF THERE IS AN OXIME ANYWHERE THEN SUBTRACT 30
```

The underlying purpose of CHMTRN is to manipulate sets programmatically and this is apparent in its vocabulary and syntax. It is neither practical nor appropriate to illustrate the use of every term in the language, or all the ways in which they can be combined a transform. The terms currently used in our SAVI project, together with brief notes about them, are listed in Tables 2 to 19. The original CHMTRN language contains many further terms – e.g. to call external subroutines that calculate a variety of physico-chemical properties such as HOMO*COEFFICIENT; ELECTROPHILIC*LOCALIZATION*ENERGY.

The original CHMTRN language used upper case characters. Our implementation is not case sensitive but, for consistency and backward compatibility, we have continued to write CHMTRN code in upper case.

The following paragraphs describe some features of the language that need more explanation than is provided in the tables.

Names of Transforms and Other Textual Values

As mentioned earlier, the first line of a CHMTRN transform contains its unique, integer identifier. The second line contains its name, entered as free text, which is intended to tell a chemist what reaction the transform describes. Examples are:

```
NAME Aldol Reaction
```

```
and
```

NAME Ester or Amide or Thiolester Formation.

Within the body of a transform, the name to be displayed to an end-user can be changed using TITLE. Actual conditions to be displayed to the user can be specified using ACTUAL*CONDITIONS (see also the section entitled “Conditions”) and warning messages can be communicated to the end user using WARNING. The statements can be conditional. For example:

TITLE 3178: Lactone Formation

IF BIT 1 IS ON THEN ACTUAL*CONDITIONS 877: Pd/Ferrocene complex, Cs₂CO₃.

For reasons of history, in all three cases, although the free text is included in the CHMTRN statement, the integer preceding it must be unique to that text. In a future implementation it might be better for the text to be maintained in separate lookup files and for only the integer identifiers to be included in the transform statements.

Subroutines

Subroutines can be called from within transforms using

CALL aaa

where aaa is the name of the subroutine and the header of the subroutine in the CHMTRN file is

SUBROUTINE aaa

Up to five atoms and bonds can be passed as parameters to the subroutine. For example,

CALL MYSUB AT ATOM*2 AND ATOM*4 AND BOND*3

Within the subroutine, the atoms and/or bonds are identified as SPECIFIED*ATOM n or SPECIFIED*BOND n, where n is determined by the position of the atom or bond in the

parameter list. So in the above example, SPECIFIED*ATOM 1 is ATOM*2, SPECIFIED*ATOM 2 is ATOM*4, and SPECIFIED*BOND 1 is BOND*3.

A subroutine terminates when a RETURN statement is encountered, and it returns SUCCESS (true) or FAIL (false) to the calling transform.

Control Statements

Simple iteration can be performed by using FOR EACH in a single statement. For example,
RAISE*RATING SLIGHTLY FOR EACH ENOLIZABLE ATOM BETA TO ATOM*2

Multiple sets can be selected for iteration by adding further EACH clauses, using “AND:FOR” for readability:

LOWER*RATING SLIGHTLY FOR EACH QUATERNARY*CENTER WITHIN &
GAMMA TO ATOM*2 AND:FOR EACH ALKYL*ATOM ALPHA TO ATOM*7

The use of DO and ENDDO in combination with FOR EACH allows multiple statements to be executed. For example,

```
FOR EACH IODINE ATOM ANYWHERE DO I_CHK
    IF THERE IS ONLY ONE ATOM ALPHA TO SPECIFIED*ATOM 1 &
        AND:IF THERE IS AN AROMATIC ATOM ALPHA &
            TO SPECIFIED*ATOM 1 THEN KILL
I_CHK ENDDO
```

As mentioned above, the terms “SPECIFIED*ATOM(S) n” and ”SPECIFIED*BOND(S) n” are used to identify parameters passed to a subroutine. As there is only one parameter passed to a DO loop in each iteration it is always SPECIFIED*ATOM 1 or SPECIFIED*BOND 1.

FOR EACH ... DO loops can be nested to multiple levels.

The terms, LOOP and JUMP, are used in the original CHMTRN language in contexts that are not relevant to SAVI and they have not currently been re-implemented. For backward compatibility, the terms, BRANCH and THEN*TO (e.g. “BRANCH TO BR_MECH THEN*TO CL_MECH”) are used only within mechanism statements (*q.v.*), although our implementation allows them to be used anywhere.

If an END*TRANSFORM statement is encountered in a branch, the retro-reaction defined by the branch is stored and displayed, and the program moves to the next branch.

Conditional Statements

Basic conditional statements take the form of a single statement in either of two formats,

yyy IF xxx

or

IF xxx THEN yyy

in which xxx is a single condition and yyy is a single outcome.

For example,

LOWER*RATING STRONGLY IF THERE IS A KETONE ALPHA TO ATOM*3

or

IF THERE IS A KETONE ALPHA TO ATOM*3 THEN LOWER*RATING STRONGLY

Depending on the design of the CHMTRN interpreter, there may be restrictions on the use of the first of these options: if the same bits in a computer word are used to store different information depending on context, “THEN” is needed to trigger separation of the statement into separate words.

Alternative conditions, and alternative second terms in a condition, can be included in a single statement using “OR:”, “AND:”, “OR:IF”, and “AND:IF”. For control statements only, an alternative outcome can be added using “OTHERWISE”. For example,

```
IF ON ATOM*3 THERE IS A KETONE OR: NITRILE AND:IF BOND*5 &  
    IS NOT A DOUBLE*BOND THEN LOWER*RATING STRONGLY &  
    OTHERWISE GOTO DB_CHK
```

The first part of a statement like, “IF ON ATOM*3 THERE IS A KETONE OR: NITRILE...”, is interpreted as meaning “IF ON ATOM*3 THERE IS A KETONE OR:IF ON ATOM*3 THERE IS A NITRILE...” Multiple instances of either AND:IF or OR:IF in the same statement are supported but the use of both in the same statement is not, to avoid ambiguity (the use of parentheses in statements of this kind is not supported).

The term “AND*THEN” allows multiple outcomes to be triggered by the same condition within a single statement. For example,

```
IF THERE IS A NITRILE ALPHA TO ATOM*3 THEN &  
    LOWER*RATING STRONGLY AND*THEN &  
    WARNING 32:Evolution of hydrogen cyanide is likely.
```

Multiple statements can be applied conditionally by using the terms BEGIN and BLKEND. For example,

```
IF THERE IS A CARBON ATOM ALPHA TO ATOM*2 OFFPATH  
BEGIN BADSIDEGRP  
  
    SAVE AS 1 THE CARBON ATOM ALPHA TO ATOM*2 OFFPATH  
  
    KILL IF THERE IS MORE THAN ONE HETERO ATOM ALPHA &  
        TO SAVED*ATOM 1
```

BLKEND BADSIDEGRP

in which the use of the label, BADSIDEGRP, is unique within the file containing the transform.

Processing within a block can be terminated, to continue with the next statement after the block end, by using EXIT followed by the label that identifies the block end. For example,

EXIT BADSIDEGRP

Mechanism Statements

Transforms include a block of code setting out the changes needed to convert the product (target) into the precursors, described as the transform mechanism and preceded and followed by a set of four dots (...). For example,

....

BREAK BOND*2

ATTACH A BROMIDE TO ATOM*3

....

The inclusion of precursor patterns and atom-to-atom correspondences in NEW*1D*PATTERNs might allow the CACTVS CHMTRN compiler to use transforms retrosynthetically or synthetically without the need for mechanism statements. However, mechanism statements are needed for backward compatibility with LHASA and there are additional issues such as stereospecificity and handling of post mechanism statements (see the following section) that currently depend on them. So the NEW*1D*PATTERNs are used for processing in the synthetic direction but the CHMTRN mechanism statements are still used for retrosynthetic processing.

Post mechanism statements

In a transform that is described retrosynthetically, statements in the body of the transform that precede the mechanism block relate to the product (target); if there are statements after the mechanism block they relate to the precursors (reactants).

Appendages

Branches in a structure can be compared using statements such as

```
KILL IF THE APPENDAGES FROM ATOM*2 TOWARDS ATOM*3 &  
AND FROM ATOM*5 TOWARDS ATOM*7 ARE NOT IDENTICAL
```

The word separators TOWARDS and AND must always be used with the term APPENDAGES.

Alternatively, where appropriate, a statement such as

```
RAISE*RATING SLIGHTLY IF ATOM*2 IS EQUIVALENT TO ATOM*5
```

tests whether the two atoms belong to the same symmetry class.

Conditions

There are two kinds of statements in CHMTRN about reaction conditions. LHASA uses a CONDITIONS statement to check for potential reactions at sites other than the intended one. It highlights the problem sites automatically to the user and may propose appropriate protection. Incompatibilities between functional groups listed in Table 7 and reaction conditions are stored in a lookup matrix. For this reason, CONDITIONS can only be selected from a predefined set.

If the actual conditions are not included in the predefined set, the transform writer selects the nearest one(s), in order to trigger automatic recognition of incompatibilities in a query structure. Multiple conditions can be listed, separated by AND. Sequences of conditions can be included, using FOLLOWED BY, to take account of changes of conditions during the course of a reaction or during workup. For example,

CONDITIONS DCCD AND R3N FOLLOWED BY pH2:4

Actual conditions are recorded in a transform using ACTUAL*CONDITIONS so that they can be reported to an end-user. They are not used by the program. Each actual conditions phrase, which is free text, must be preceded by a unique identifying number and a colon. For example,

IF BIT 1 IS ON THEN ACTUAL*CONDITIONS 878: Pd/BINAP, Cs2CO3

Rating

The knowledge base for the LHASA program originally included a numerical rating in the range 0 to 100, in increments of 5, for each transform, which was the transform writer's assessment of the utility of the retro-reaction it described. The synonymous terms, RATING and INITIAL*RATING are supported. For example,

RATING 65

or

INITIAL*RATING 65

The rating value is not the likely percentage yield of the reaction: it takes into account factors such as how well-documented a reaction is and how consistent yields are (one reaction might deliver only 40% yields but do so consistently, while another might deliver yields ranging from a few percent to almost one hundred percent with no clear reason for the variability). Within the

body of a transform the initial rating can be increased or decreased by multiples of five using ADD (or INCREMENT) or SUBTRACT (or DECREMENT). For example,

ADD 25 IF THERE IS A DONATING GROUP ON ATOM*3

It was difficult to set out guidelines for consistent initial rating of transforms by different writers. So an alternative approach was introduced in the 1990s, in which the different factors were assessed independently. In addition, instead of using numerical values, the individual factors were assigned categorical values. An initial rating was calculated automatically by the program, allowing the weighting of the different factors to be adjusted. The factors contributing to rating were as follows. In each case guidelines can be drawn up on how to decide upon the appropriate category (e.g. what average yield should be considered GOOD and what FAIR). As well as potentially improving consistency, separating the factors contributing to the rating makes it possible for an end-user to customise priorities if, say, cost is a primary consideration.

CONDITION*FLEXIBILITY

An assessment of how many different types of reagents can be used for the reaction.

HETEROSELECTIVITY

An assessment of how selective the reaction is towards the functional groups in the reaction centre compared with other functional groups.

HOMOSELECTIVITY

An assessment of how selective the reaction is towards the functional group(s) in the intended site of reaction compared with instances of the same functional group(s) in other environments in the same structure.

ORIENTATIONAL*SELECTIVITY

An assessment of the selectivity of the reaction towards one location in the site of reaction compared with another (e.g. leading to Markovnikov or anti-Markovnikov addition).

RELIABILITY

An assessment of how predictable a reaction is, both in terms of yield and, where relevant, stereochemical outcome.

REPUTATION

How widely reported, and reproduced, the reaction is (or, to be more exact, was at the time that the transform was written or last updated).

THERMODYNAMICS

How easily the reaction proceeds spontaneously.

TYPICAL*YIELD

A category based on the percentage yield normally to be expected.

More recently, we have added the following two factors.

COST

Whether expensive reagents or costly conditions, such as use of high temperature and pressure, are needed.

SAFETY

Whether hazardous reagents or conditions are involved.

Categories that can be assigned to rating factors are: BAD, POOR, FAIR, GOOD, EXCELLENT, and NOT*APPLICABLE. In addition, for TYPICAL*YIELD only, the category VERY*GOOD can be used.

Qualitative rating adjustments

Although the scheme for qualitative assignment of initial rating described above was implemented for LHASA, increments and decrements within the body of transforms remained numerical. We have introduced an alternative, qualitative categorical scheme, allowing all rating assessments to be under the control of the application that uses the transform knowledge base, where it can be done by reasoning about the relationships between categories or arithmetically.

The words RAISE*RATING and LOWER*RATING have been introduced along with terms to express the degree of change that is required: SLIGHTLY, MODERATELY, STRONGLY, or SEVERELY. For example:

IF THERE IS AN ALCOHOL OFFPATH THEN LOWER*RATING STRONGLY
RAISE*RATING SLIGHTLY IF BOND*2 IS IN A RING OF SIZE 5 THROUGH 7

For backward compatibility, numerical rating increments and decrements are still being written into transforms as well. For example, the above statement about lowering rating if there is an alcohol offpath might be accompanied by:

IF THERE IS AN ALCOHOL OFFPATH THEN SUBTRACT 30

Guidelines for deciding on the degree of change to make to a rating have not yet been fully developed and there were previously no clear guidelines for deciding on the size of numerical increments or decrements. The broad advice on numerical increments and decrements was to take into consideration the size of the initial numerical rating for the transform, whereas the new scheme requires only a judgement about the change in itself – scaling of the change to suit the initial, categorical rating, where appropriate, is handled by the program (and potentially the end user) rather than by the transform writer. So there is no formal mapping between numerical and

categorical rating changes. For the purposes of updating old transforms to the new scheme, after looking at how numerical rating changes (which were always in multiples of five) had been used in practice, we have applied the following conversions of numerical to categorical values: 5-15, SLIGHTLY; 20-25, MODERATELY; 30-35, STRONGLY; ≥ 40 , SEVERELY.

Although, as mentioned above, our transforms currently contain rating statements of both kinds, a CHMTRN interpreter should use only one rating scheme or the other, of course.

Stereochemistry

The terms SYN and ANTI, with the meanings understood by a chemist, are used to describe double bond geometry and also diastereoisomerism. They can be used in relation to atoms and/or bonds.

In the case of double bonds, the usage is straightforward. For example,

IF ATOM*1 IS ANTI TO ATOM*4 ...

IF ATOM*1 IS ANTI TO BOND*3 ...

Defining the relationship between atoms or bonds on a ring or saturated chain is more complicated and uses the approach described by Masamune *et al.*,¹³ using an additional atom attached to the same atom as each of those of interest to define the layout of the path between them. For example,

IF ATOM*1 IS SYN TO ATOM*5 ABOUT THE PATH FROM ATOM*2 TO ATOM*6 ...

might be used to test the stereochemical relationship between atom A1 and atom A5 in Figure 5. The chain between atom A2 and atom A6 is laid out zigzag fashion as shown in the figure. If A1 and A5 are both above, or both below, the plane of the paper, then they are recognised as SYN to each other. Otherwise they are ANTI.



Figure 5. Layout of a structure to define a stereochemical relationship

Properties Relating to Rings

The original CHMTRN includes a rich set of terms relating to the properties of atoms and bonds in rings which are not currently used in SAVI. For this reason the following terms have not yet been re-implemented: 3*OR*MORE*RINGS, CIS*BRIDGE(D), CORE, LOCKED*BOAT, LOCKED*CHAIR, OFF*THE*BRIDGE, ON*THE*BRIDGE, SPIRO, TRANS*BRIDGE(D).

Strained Bonds

A bond is considered strained if it is:

in a three- or four-membered ring;

or in a five-membered ring that is part of a [2.1.1], [2.2.1], or [3.2.1] system;

or a trans double bond in a ring of size 9 or less;

or a double bond exocyclic to a three- or four-membered ring.

The atom(s) attached to the bond and in the ring are also classified as strained and so statements like either of the following can be used:

IF BOND*3 IS STRAINED ...

IF ATOM*5 IS STRAINED ...

The term BREDT*STRAINED refers to bridgehead double bonds that are in a ring of size less than 8 and to the associated bridgehead atom:

IF ATOM*1 IS BREDT*STRAINED ...

AN EXAMPLE TRANSFORM

Figure 6 illustrates the use of CHMTRN, with embedded PATRAN, in a transform. The retro-reaction diagram following the references is a representation for the convenience of anyone working on the transform – it is ignored by the compiler. As mentioned above, there are statements to modify the rating both numerically (SUBTRACT 40), for backward compatibility, and qualitatively (LOWER*RATING SEVERELY). The mechanism statements encode the separation of R¹ from the nitrogen atom and the attachment of a B(OH)₂ group to R¹.

CONCLUSION

We have been using the revised CHMTRN for CACTVS-assisted generation of compound libraries successfully based on a preliminary set of 79 widely-applicable transforms. As synthetic chemistry keeps developing, it is easy to write new transforms in CHMTRN, providing a powerful tool for defining and exploring the chemical universe. Our hope is that by illustrating the power and advantages offered by CHMTRN we can encourage the wider use that it deserves.

TRANSFORM 7022
 NAME Chan-Lam coupling
 VERSION 1
 WRITTEN*BY P. Judson, 2018-12-17
 CHANGED*BY W. Ihlenfeldt, 2019-07-08

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END*REFERENCES

TYPICAL*YIELD	GOOD
RELIABILITY	GOOD
REPUTATION	GOOD
HOMOSELECTIVITY	POOR
HETEROSELECTIVITY	FAIR
ORIENTATIONAL*SELECTIVITY	NOT*APPLICABLE
CONDITION*FLEXIBILITY	POOR
THERMODYNAMICS	GOOD

1D*PATTERN

C[ARYL=YES]-N,O,S-C

1D*PATTERN

C[FGS=OLEFIN]-N,O,S-C

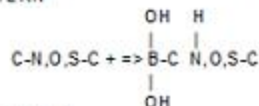
NEW*1D*PATTERN

C[ARYL=YES]-N,O,S-C => C^1[ARYL=YES]-B(-O[HS=1])-O[HS=1] + N,O,S^2[HS>0]-C^3

NEW*1D*PATTERN

C[ARYL=YES]-N,O,S-C => C^1[FGS=OLEFIN]-B(-O[HS=1])-O[HS=1] + N,O,S^2[HS>0]-C^3

2D*PATTERN



END*PATTERNS

DISCONNECTIVE
 BROKEN*BONDS BOND*1

...
 KILL IF ANYWHERE OFFPATH THERE IS AN ALCOHOL OR: THIOL GROUP
 FOR EACH NITROGEN ATOM OFFPATH DO N_CHK
 KILL IF THERE IS A HYDROGEN ON SPECIFIED*ATOM 1

N_CHK ENDDO

...The reaction works with, e.g., ureas and carbamates but there are no
 ...examples of reactions with, e.g., carboxylic acids. The reaction
 ...appears to be so versatile that it is difficult to come up with any
 ...other restrictions.
 IF THERE IS MORE THAN ONE HETERO ATOM ALPHA TO ATOM*3 AND:IF &
 ATOM*2 IS NOT NITROGEN THEN SUBTRACT 40
 IF THERE IS MORE THAN ONE HETERO ATOM ALPHA TO ATOM*3 AND:IF &
 ATOM*2 IS NOT NITROGEN THEN LOWER*-rating SEVERELY

...
 CONDITIONS Cu[I]/Pyr
 ACTUAL*CONDITIONS 913: Cu(OAc)2/Et3N/air/4 Angstrom mol. sieve

...
 BREAK BOND*1
 INTRODUCE A BORON ATOM ON ATOM*1
 KEEP THE NEWEST*ATOM
 INTRODUCE AN OXYGEN ON THE KEPT*ORIGIN
 INTRODUCE AN OXYGEN ON THE KEPT*ORIGIN

...

Figure 6. An example of a transform

AUTHOR INFORMATION

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Competing interests

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Table 1. Atom and Bond Properties Supported in 1D Patterns

Code	Property	Allowed values
ARYL	Whether an atom is aromatic	YES, NO, EITHER
CHARGE	Whether an atom is charged or a radical	ANION, CATION, NEUTRAL ^a , RADICAL, YES, NO ^a , EITHER
EPSb	The number of electron pairs on an atom	0, 1, 2, 3, 4
FGNOT	Functional groups that are not allowed	Any of the groups listed in Table 7
FGS	Functional groups one of which must be present	Any of the groups listed in Table 7
HETSb	The number of hetero atoms alpha to an atom	0, 1, 2, 3, 4
HSb	The number of hydrogens (explicit and implicit) on an atom	0, 1, 2, 3, 4
RINGb	The size of a ring containing an atom	0 or an integer of 3 or more, YES, NO ^c , EITHER ^d
FUSIONE	Whether a bond is a fusion bond between rings	DIARYL, ALKYLARYL, DIALKYL, TRIALKYL, YES, NO, EITHER

- a) NEUTRAL and NO are synonymous.
- b) In addition to =, the operators > and < are supported for numerical values.
- c) RING=0 means the same as RING=NO (i.e. the atom is acyclic).
- d) EITHER means the atom is allowed to be cyclic or acyclic.
- e) In addition to =, the operator # is supported, representing “not equals”.

Table 2. Transform Properties

Reserved word ^a	Notes
1D*PATTERN	See the section entitled ‘Patterns’
2D*PATTERN	See the section entitled ‘Patterns’
ACTUAL*CONDITIONS n:aaa	See the section entitled ‘Conditions’
CONDITION*FLEXIBILITY aaa	See the section entitled ‘Rating’
CONDITIONS aaa	The most similar reaction conditions in a lookup matrix to the actual ones used for the reaction that a transform describes. The matrix contains information on the sensitivity of functional groups to reaction conditions and supports automatic attachment of protecting groups in LHASA. See the section entitled ‘Conditions’.
COPYRIGHT aaa	Copyright claim (e.g. “COPYRIGHT Smith and Co., 2019”)
COST aaa	See the section entitled ‘Rating’
END*PATTERN(S)	See the section entitled ‘Patterns’
END*REFERENCES	Marks the end of the section in a transform containing bibliographic references and similar information (see REFERENCES).
HETEROSELECTIVITY aaa	See the section entitled ‘Rating’
HOMOSELECTIVITY aaa	See the section entitled ‘Rating’
INITIAL*RATING n	See the section entitled ‘Rating’
NAME aaa	The name of the transform (e.g. “NAME Aldol Reaction”). Transform names do not need to be unique (the unique identifier is the transform number).
NEW*1D*PATTERN	See the section entitled ‘Patterns’
ORIENTATIONAL*SELECTIVITY aaa	See the section entitled ‘Rating’
OPT*REFERENCES aaa	Identifies optional reference(s), separated by “AND”, that are to be displayed to the user (e.g. “IF THERE IS A KETONE GROUP ON ATOM*1 THEN OPT*REFERENCES 3 AND 5”).

OPTIONAL*REFERENCES	Marks the start of an optional subsection at the end of the REFERENCES section, containing bibliographic references that are only displayed to the user if they are selected by a statement in the main body of the transform. Each reference in the optional references subsection is preceded by “nnn: ”, where nnn is an integer that is unique in the optional references list of the transform.
RATING n	See the section entitled ‘Rating’
REFERENCES	Marks the beginning of the section in a transform containing bibliographic references and similar information such as a brief textual or diagramatic description of the reaction/retro-reaction.
RELIABILITY aaa	See the section entitled ‘Rating’
REPUTATION aaa	See the section entitled ‘Rating’
SAFETY aaa	See the section entitled ‘Rating’
THERMODYNAMICS aaa	See the section entitled ‘Rating’
TITLE n:aaa	An alternative transform name, specified in the main body of the transform, to be displayed to the user in place of the one set by NAME.
TRANSFORM n	The unique integer identifier for the transform.
TYPICAL*YIELD aaa	See the section entitled ‘Rating’

a) aaa represents a string value; n represents an integer.

Table 3. Rating Variables

Reserved word
MODERATELY
SEVERELY

SLIGHTLY
STRONGLY
BAD
POOR
FAIR
GOOD
VERY*GOOD
EXCELLENT
NOT*APPLICABLE

Table 4. Atom Type

ALUMINIUM^a
BORON
BROMINE(S)
CARBON(S)
CHLORINE(S)
COPPER
FLUORINE(S)
HYDROGEN*ATOM(S)^b
IODINE(S)
NITROGEN(S)
OXYGEN(S)
PHOSPHORUS
SELENIUM
SILICON(S)
SULFUR

SULPHUR

STANNUM

TIN

TITANIUM

a) The use of “aluminum” as an alternative IUPAC name was not formally adopted until 1993, a long time after CHMTRN was created. It could be added, in the same way as SULFUR and SULPHUR are supported, but we have not so far needed to do so.

b) See the section entitled ‘Hydrogen atoms’

Table 5. Other Atom Properties

Reserved word ^a	Notes ^b
ALKALI*METAL	The atom is Li, Na, K, Rb, or Cs
ALKYL(S)	The number of ALKYL*ATOMs (see Table 8) attached to the atom. The return value is an integer.
ALLYLIC	The atom is a saturated carbon atom alpha to a carbon-carbon double bond
ANTI	1) Tests the relationship between a pair of atoms and/or bonds about a double bond. 2) Tests the diastereomeric relationship between a pair of atoms and/or bonds. See the section entitled ‘Stereochemistry’.
AROMATIC	The atom is aromatic.
ARYL	Synonym for AROMATIC.
BENZYLIC	A saturated carbon atom that is alpha to an aromatic carbon atom.
BREDT*STRAINED	The atom is on a double bond which is at a bridgehead and in a ring of size less than eight. See the section entitled ‘Strained Bonds’.

BRIDGEHEAD(S)	Bridgehead atoms in bridged cyclic systems.
BRIDGED*TO*CURRENT*RING	The atom is in a chain that bridges the CURRENT*RING.
CIS	Tests whether two atoms on the same ring are cis to each other. The two atoms and the ring must all be specified (e.g. "IF ATOM*1 IS CIS TO ATOM*2 ABOUT THE RING CONTAINING BOND*4 THEN...")
CIS*OLEFIN	A carbon atom joined to another carbon atom by a double bond where both carbon atoms carry one (and only one) hydrogen atom and the substituents are in a cis relationship to each other.
COMMON	Tests whether atoms and/or bonds are in the same ring (e.g. "IF ATOM*1 AND BOND*3 ARE IN A COMMON RING ...").
CONJUGATED	An atom that is alpha to a multiply-bonded atom which is connected to, but not part of, a structural group that keys the transform.
DOUBLY	There is a double bond on the atom (normally used with the buzz word BONDED – i.e. "DOUBLY BONDED").
ELECTRON*DENSITY	The electron density calculated for the atom (the return value is decimal).
ENOLIZABLE	The atom is saturated, not a bridgehead atom, has at least one hydrogen atom on it, and is activated by a withdrawing bond.
EQUIVALENT	Tests whether two atoms belong to the same symmetry class (e.g. "RAISE*RATING MODERATELY IF ATOM*3 IS EQUIVALENT TO ATOM*4"). See the section entitled 'Appendages'.
EXPLICIT*HYDROGEN	See the section entitled 'Hydrogen atoms'.
FUSED	The atom is at a ring fusion.
FUSED*TO*CURRENT*RING	The atom is in a ring that is fused to the CURRENT*RING.
FUSION	Synonym for FUSED.
HALOGEN*ATOM(S)	The atom(s) is/are F, Cl, Br, or I.

HETERO	The atom is an element other than carbon or hydrogen.
HYDROGEN(S)	The number of hydrogen atoms (explicit and implicit) attached to the atom (the return value is an integer).
LESS*HINDERED	Compares the steric hindrance at two atoms in terms of the number of non-hydrogen atoms attached to each (e.g. "IF ATOM*1 IS LESS*HINDERED THAN ATOM*3 THEN ...").
LESS*HINDERED THAN	Synonym for LESS*HINDERED.
LESS*HINDERED*THAN	Synonym for LESS*HINDERED.
METHYL(S)	The number of METHYL*ATOM(S) (i.e. CH ₃ groups) attached to the atom (the return value is an integer).
MORE*HINDERED	Compares the steric hindrance at two atoms in terms of the number of non-hydrogen atoms attached to each (e.g. "IF ATOM*1 IS MORE*HINDERED THAN ATOM*3 THEN ...").
MORE*HINDERED THAN	Synonym for MORE*HINDERED.
MORE*HINDERED*THAN	Synonym for MORE*HINDERED.
MULTIPLY	A bond on the atom is double, triple, or aromatic.
MULTIPLY*BONDED	Synonym for MULTIPLY.
NOT*ANTI	Negates the value returned for ANTI (q.v.)
NOT*SYN	Negates the value returned for SYN (q.v.)
OFF*CURRENT*RING	The atom is not in the ring that has been designated as the CURRENT*RING.
OFF*THE*RING	Synonym for OFF*CURRENT*RING.
OFFPATH	The atom is not included in the keying pattern for the transform.
OFF*PATH	Synonym for OFFPATH.
OFFRING	Synonym for OFF*CURRENT*RING.
OFF*RING	Synonym for OFF*CURRENT*RING.
OFF*THE*BRIDGE	See the section entitled 'Bridged Rings'.
ON*THE*BRIDGE	See the section entitled 'Bridged Rings'.

ON*THE*RING	Synonym for ON*CURRENT*RING.
ONPATH	The atom is included in the keying pattern for the transform.
ON*PATH	Synonym for ONPATH.
ON*CURRENT*RING	The atom is in the ring that has been designated as the CURRENT*RING.
ON*RING	Synonym for ON*CURRENT*RING.
ONRING	Synonym for ON*CURRENT*RING.
POSITIVE*CHARGE	The atom carries a positive charge.
PRIMARY*CENTER(S)	There is one (and only one) carbon atom attached to the atom.
PRIMARY*CENTRE(S)	Synonym for PRIMARY*CENTER(S).
QUATERNARY*CENTER(S)	There are four carbon atoms attached to the atom.
QUATERNARY*CENTRE(S)	Synonym for QUATERNARY*CENTER(S).
SECONDARY*CENTER(S)	There are two (and only two) carbon atoms attached to the atom.
SECONDARY*CENTRE(S)	Synonym for SECONDARY*CENTER(S).
STRAINED	A bond on the atom is STRAINED. See the section entitled 'Strained Bonds'.
SYN	1) Tests the relationship between a pair of atoms and/or bonds about a double bond. 2) Tests the diastereomeric relationship between a pair of atoms and/or bonds. See the section entitled 'Stereochemistry'.
TERTIARY*CENTER(S)	There are three (and only three) carbon atoms attached to the atom.
TERTIARY*CENTRE(S)	Synonym for TERTIARY*CENTER(S).
TRANS	Tests whether two atoms on the same ring are trans to each other. The two atoms and the ring must all be specified (e.g. "IF ATOM*1 IS TRANS TO ATOM*2 ABOUT THE RING CONTAINING BOND*4 THEN...")

TRANS*OLEFIN	A carbon atom joined to another carbon atom by a double bond where both carbon atoms carry one (and only one) hydrogen atom and the substituents are in a trans relationship to each other.
--------------	---

TRIPLY	There is a triple bond on the atom (normally used with the buzz word BONDED – i.e. “TRIPLY BONDED”).
--------	--

a) Where an optional letter S is shown, either word can be used synonymously. E.g. “ALKYL” and “ALKYLS” are both supported (the combined term, “ALKYL(S)”, is used for convenience in the table and is not part of the language).

b) The values of properties are Boolean (true or false) unless otherwise indicated.

Table 6. Bond Properties

Reserved word	Notes
ALLYLIC	The bond is on an allylic carbon atom and is not the bond linking that atom to the doubly bonded one.
ANTI	1) Tests the relationship between a pair of atoms and/or bonds about a double bond. 2) Tests the diastereomeric relationship between a pair of atoms and/or bonds. See the section entitled ‘Stereochemistry’.
AROMATIC	The bond is aromatic.
ARYL	Synonym for AROMATIC.
BREDT*STRAINED	The bond is a double bond at a bridgehead and in a ring of size less than eight. See the section entitled ‘Strained Bonds’.
BRIDGEHEAD(S)	The bonds attached to bridgehead atoms in bridged cyclic systems.
BRIDGED*TO*CURRENT*RING	The bond is in a chain that bridges the CURRENT*RING.
CIS*OLEFIN	A double bond between two carbon atoms where both carbon atoms carry one (and only one) hydrogen

	atom and the substituents are in a cis relationship to each other.
COMMON	Tests whether atoms and/or bonds are in the same ring (e.g. "IF BOND*1 AND ATOM*3 ARE IN A COMMON RING ...").
CONJUGATED	A bond on an atom that is alpha to a multiply-bonded atom in a functional group and is not the one connecting the atom to the functional group.
DIARYL*FUSION	A fusion bond between two aromatic rings.
DONATING	The bond is activated/deactivated by a DONATING group.
DOUBLE*BOND(S)	A double bond.
FUSED*TO*CURRENT*RING	The bond is in a ring that is fused to the CURRENT*RING.
FUSION	A fusion bond between two rings.
MULTIPLE	A double, triple, or aromatic bond.
MULTIPLE*BOND(S)	Synonym for MULTIPLE.
NOT*ANTI	Negates the value returned for ANTI (q.v.).
NOT*SYN	Negates the value returned for SYN (q.v.).
OFF*CURRENT*RING	The bond is not in the ring that has been designated as the CURRENT*RING.
OFFPATH	The bond is not included in the keying pattern for the transform.
OFFRING	Synonym for OFF*CURRENT*RING.
OFF*SPECIFIED*PATH	Used only in subroutines. The bond is not included in the set passed to the subroutine as part of the specified path.
ON*CURRENT*RING	The bond is in the ring that has been designated as the CURRENT*RING.
ONPATH	The bond is included in the keying pattern for the transform.
ON*SPECIFIED*PATH	Used only in subroutines. The bond is included in the set passed to the subroutine as part of the

	specified path.
ONRING	Synonym for ON*CURRENT*RING.
SINGLE*BOND(S)	A single bond.
SMALL*FUSION	A fusion bond between a pair of rings of sizes 3/3, 3/4, 3/5, 3/6, 4/4, 4/5, 4/6, or 5/5.
SMALL*FUSIONS*BOND(S)	Synonym for SMALL*FUSION.
STRAINED	See the section entitled 'Strained Bonds'.
SYN	1) Tests the relationship between a pair of atoms and/or bonds about a double bond. 2) Tests the diastereomeric relationship between a pair of atoms and/or bonds. See the section entitled 'Stereochemistry'.
TRANS*OLEFIN	A double bond between two carbon atoms where both carbon atoms carry one (and only one) hydrogen atom and the substituents are in a trans relationship to each other.
TRIPLE*BOND(S)	A triple bond.
WITHDRAWING	The bond is activated/deactivated by a WITHDRAWING group. Note that the withdrawing bonds of a non-expandable withdrawing group (e.g. NITRO) are on the same atom as the withdrawing group; the withdrawing bonds of an expandable withdrawing group (e.g. KETONE) are on the atom(s) alpha to it.

Table 7. Functional Groups

Reserved word ^a	Substructure ^b
ACETAL ^c	C-O-C*-O-C
ACETYLENE	C*≡C
ACID	C*(=O)-OH
ACID*HALIDE	C*(=O)X [X = Cl, Br, or I]

ALCOHOL	C^*-OH
ALDEHYDE	$C^*(=O)-H$
ALLENE	$C^*=C=C$
AMIDE	AMIDE*1, AMIDE*2, or AMIDE*3
AMIDE*1	$C^*(=O)-NH_2$
AMIDE*2	$C^*(=O)-NH-C$
AMIDE*3	$C^*(=O)-N(-C)-C$
AMIDZ	$C^*-N-C=O$
AMINE	AMINE*1, AMINE*2, or AMINE*3
AMINE*1	C^*-NH_2
AMINE*2	C^*-NH-C
AMINE*3	$C^*-N(-C)-C$
AMINE*OXIDE	$C^*-N^+(-O^-)(-C)-C$
ANHYDRIDE	$C^*(=O)-O-C=O$
AZIDE	$C^*-N=N^+=N^-$
AZIRIDINE	$C^* - N$
	$\backslash /$
	C
AZO	$C^*-N=N-C$
BROMIDE	C^*-Br
C*SULFONATE	C^*-SO_2-O
C*SULPHONATE	Synonym for C*SULFONATE
CARBAMATE*C	$N(-C)(-C)-C^*(=O)-O-C$
CARBAMATE*H	$NH-C^*(=O)-O$
CARBONIUM	C^+
CARBONYL	ALDEHYDE or KETONE
CARBOXYL	ACID, ESTER, AMIDE, ACID*HALIDE,

	THIOESTER, or ANHYDRIDE
CHLORIDE	C^*-Cl
CYANO	$C^*\equiv N$
DIAZO	$C^*=N^+=N^-$
DISULFIDE	$C^*-S-S-C$
DISULPHIDE	Synonym for DISULFIDE
DITHIOACETAL	$RS-C^*H-SR'$
DITHIOKETAL	$RS-C^*(-C)(-C)-SR'$
ENAMINE	$C=C^*-N(-C)-C$
ENOL*ETHER	$C=C^*-O-C$
EPISULFIDE	C^*-S
	$\backslash /$
	C
EPISULPHIDE	Synonym for EPISULFIDE
EPOXIDE	C^*-O
	$\backslash /$
	C
ESTER	$C^*(=O)-O-C$
ESTERX	$C^*-O-C=O$
ETHER	C^*-O-C
FLUORIDE	C^*-F
FUNCTIONAL	Any of the groups defined in this table
GEM*DIHALIDE	C^*XY [X, Y = F, Cl, Br, or I]
GLYCOL	$R-O-C^*-C-O-R'$ [R, R' = C or H]
HALIDE	BROMIDE, CHLORIDE, FLUORIDE, or IODIDE
HALOAMINE	C^*-N-X [X = F, Cl, Br, or I]

HALOHYDRIN	$X-C^*-O-R$ [X = F, Cl, Br, or I; R = C or H]
HEMIACETAL	$C-O-C^*H-OH$
HYDRATE	$HO-C^*-OH$
HYDRAZONE	$C^*=N-N$
HYDROXYL	Synonym for ALCOHOL
HYDROXYLAMINE	C^*-N-OH
IMINE	$C^*=N-R$ [R = C or H]
IODIDE	C^*-I
ISOCYANATE	$C^*-N=C=O$
ISOCYANIDE	$C^*-N^+\equiv C^-$
KETONE	$C-C^*(=O)-C$
LACTAM	$O=C^*-NH$
	$\backslash /$
	C_n
LACTONE	$O=C^*-O$
	$\backslash /$
	C_n
METHYLENE	$C^*=CH_2$
N*CARBAMATE	$C^*-N-C(=O)-O-C$
N*UREA*C	$C^*-N(-C)-C(=O)-N(-C)-C$
N*UREA*H	$C^*-N-C(=O)-NH$ or $C^*-NH-C(=O)-N$
NITRILE	Synonym for CYANO
NITRITE	$C^*-O-N=O$
NITRO	$C^*-N^+(=O)-O^-$
NITROSO	$C^*-N=O$
O*CARBAMATE	$C^*-O-C(=O)-N$
O*CARBONATE	$C^*-O-C(=O)-O-C$

O*SULFONATE	C^*-O-SO_2-R [R = C or H]
O*SULPHONATE	Synonym for O*SULFONATE
OLEFIN	$C^*=C$
OXIME	$C^*=N-OH$
PEROXIDE	C^*-O-OH or $C^*-O-O-C$
PHOSPHINE	$C^*-P(-C)-C$
PHOSPHONATE	C^*-O-P
SELENIDE	C^*-Se-C
SILYLENOLEATHER	$C=C^*-O-Si(-C)(-C)-C$
SULFIDE	C^*-S-C
SULFONE	C^*-SO_2-C
SULFOXIDE	$C^*-S(=O)-C$
SULPHIDE	Synonym for SULFIDE
SULPHONE	Synonym for SULFONE
SULPHOXIDE	Synonym for SULFOXIDE
THIOCYANATE	$C^*-S-C\equiv N$
THIOESTER	$C^*(=O)-S-C$
THIOL	C^*-SH
TRIALKYLSELOXY	$C^*-O-Si(-C)(-C)-C$
TRIALKYLSELYL	$C^*-Si(-C)(-C)-C$
TRIHALIDE	C^*XYZ [X, Y, Z = F, Cl, Br, or I]
VIC*DIHALIDE	$X-C^*-C-Y$ [X, Y = F, Cl, Br, or I]
VINYLSILANE	$C=C^*-Si(-C)(-C)-C$

a) Note that some group names are not unique – e.g. ALCOHOL and HYDROXYL are synonyms

b) The atom on which the group is centred is indicated with an asterisk. It is always a carbon atom.

c) Note that, the definition for ACETAL in the original CHMTRN, and currently retained, includes ketals.

Table 8. Other substructures

Reserved word	Notes
ALKYL*ATOM(S)	An sp^3 hybridised carbon atom with no bonds to heteroatoms.
APPENDAGE(S)	Allows a pair of substructures within a structure to be compared. See the section entitled 'Appendages'.
ISOPROPYL	$CH_3-CH^*-CH_3$
METHYL*ATOM(S)	A carbon atom with three hydrogens (explicit and/or implicit) attached to it.
PHENYL	$ \begin{array}{c} CH=CH \\ / \quad \backslash \\ C^* \quad CH \\ \backslash \quad // \\ CH-CH \end{array} $
T*BUTYL	$CH_3-CH^*(-CH_3)-CH_3$

Table 9. Group Properties

Reserved word	Notes ^a
DONATING	Any of ACETAL, ALCOHOL, AMIDZ, AMINE, ESTERX, ETHER, HALOAMINE, HEMIACETAL, HYDROXYLAMINE, N*CARBAMATE, N*UREA*C, N*UREA*H, O*CARBAMATE, O*CARBONATE,

	PHOSPHINE, SELENIDE, SULFIDE, THIOL, TRIALKYLSILOXY, or VINYL
EXPANDABLE*WITHDRAWING ^b	Any of ACID*HALIDE, ALDEHYDE, AMIDE, ANHYDRIDE, CARBONIUM, ESTER, KETONE, IMINE, NITRILE, OXIME, THIOESTER, TRIHALIDE, or VINYLW
GOOD*LEAVING	Any of AZIRIDINE, BROMIDE, CHLORIDE, EPISULFIDE, EPOXIDE, IODIDE, O*SULFONATE, PHOSPHONATE, GEM*DIHALIDE, or TRIHALIDE
GOOD*LEAVING*GROUP	Synonym for GOOD*LEAVING
INTERFERING	The transform writer can designate a group as one that will interfere with the reaction described by a transform and is likely to be unprotectable, so that the end-user's attention can be drawn to the problem. An automatic decrement may also be applied to the transform by the program. See also PROTECTED.
LEAVING	Any of ACETAL, ALCOHOL, AMIDZ, AZIRIDINE, CARBAMATE (N- and O-), DITHIOACETAL, DITHIOKETAL, EPISULFIDE, EPOXIDE, ESTERX, ETHER, GEM*DIHALIDE, HALIDE, N*UREA*C, N*UREA*H, O*CARBONATE, PHOSPHONATE, SELENIDE, SULFIDE, SULFONE, SULFONATE (C- and O-), THIOCYANATE, THIOL, TRIALKYLSILOXY, or TRIHALIDE
LEAVING*GROUP	Synonym for LEAVING
NON*EXPANDABLE*WITHDRAWING ^b	NITRO, C*SULFONATE, SULFONE, or SULFOXIDE
PROTECTED	The transform writer can designate a group as one that the program should treat as being protected while the reaction is carried out, so that it can be appropriately labelled in displays and does not trigger automatic decrements to the transform rating. See also INTERFERING.
PARTICIPATING	The transform writer can indicate that what might appear to be a functional group that

would compete with the reaction described by a transform is not to be taken into account when assessing the need for protection.

VINYL*D

C*=C-D [D = donating group, q.v.]

VINYL*W

C*=C-W [W = withdrawing group, q.v.]

WITHDRAWING

Any EXPANDABLE*WITHDRAWING or NONEXPANDABLE*WITHDRAWING group

- a) The values of properties are Boolean.
- b) An expandable withdrawing group is one that activates the atom(s) alpha to the origin of the group, (e.g., ketone, ester); a non-expanding withdrawing group activates the origin of the group itself (e.g. nitro). So WITHDRAWING BONDS associated with a non-expanding withdrawing group are the bonds on the origin itself (excluding those that are defined as part of the group); WITHDRAWING BONDS associated with an expandable withdrawing group are the bonds on the atom(s) alpha to the origin (excluding the atoms that are defined as part of the group).

Table 10. Atom and Bond Sets

Reserved word ^a	Notes
ALPHA	The atoms alpha to a specified atom and the bonds on them, excluding the bonds back to the specified atom (e.g. "ALPHA TO ATOM*1").
ATOMS*TO*ATOMS	The atoms alpha to a specified atom or set of atoms.
ATOMS*TO*BONDS	The bonds on a specified atom or set of atoms.
BETA	The atoms and bonds beta to a specified atom or set of atoms (e.g. "BETA TO ATOM*1").
BOND(S)	The set of bonds in a structure/substructure.
BONDS*TO*ATOMS	The atoms attached to a specified bond or set of bonds.
BONDS*TO*BONDS	The bonds on atoms attached to a specified bond or set of bonds (excluding the specified bond(s)).

BRIDGE*1	See the section entitled 'Bridged Rings'.
BRIDGE*2	See the section entitled 'Bridged Rings'.
BRIDGE*3	See the section entitled 'Bridged Rings'.
CURRENT*RING	The ring designated as current (see DESIGNATE in table 14).
FRAGMENT*1	The precursor containing the lowest numbered atom.
FRAGMENT*2	Where there are two precursors, the one not containing the lowest numbered atom.
GAMMA	The atoms and bonds gamma to a specified atom or set of atoms (e.g. "GAMMA TO ATOM*1").
IT	The atom, bond, or set of either, identified in the preceding CHMTRN statement.
LOCANT(S)	Synonym of IT. Intended only for use together with the buzz word PREVIOUS for clarity (e.g. "IF THERE IS A NITROGEN ATOM ALPHA TO THE PREVIOUS LOCANT").
RING(S)	The set of all rings in a structure/substructure. Supports statements such as "IF ATOM*1 IS IN A RING..."
SAVED*ATOM(S) n	The set of atoms saved using "SAVE AS n..."
SAVED*BOND(S) n	The set of bonds saved using "SAVE AS n..."
SPECIFIED*ATOM(S) n	A set of atoms passed to a subroutine or an atom passed to a DO loop.
SPECIFIED*BOND(S) n	A set of bonds passed to a subroutine or a bond passed to a DO loop.
SPANNING	The bond between a specified pair of atoms and/or bonds. Both atoms must be specified, separated by the word SPANNING (e.g. "IF THE BOND SPANNING ATOM*1 AND ATOM*6 IS A DOUBLE*BOND THEN ...").
THEM	Synonym of IT.
WITHIN	Used only together with ALPHA, BETA, or GAMMA. The set of all atoms and bonds within the stated distance from a specified atom, or set of atoms, excluding the atom itself and the bonds on it (e.g. "WITHIN BETA TO ATOM*1" means the union of the sets of atoms alpha and beta to ATOM*1).

a) n represents an integer.

Table 11. Properties of Sets

Reserved word ^a	Notes ^b
BRIDGE(D) ^c	See the section entitled ‘Bridged Rings’.
CARBOCYCLIC	Ring property – the ring contains only carbon atoms.
HETEROCYCLIC	Ring property – the ring contains at least one non-carbon atom.
FEWER THAN aaa	Tests the number of objects in a set (e.g. “IF THERE ARE FEWER THAN TWO HYDROGENS ON ATOM*1 ...”).
FEWER*THAN aaa	Synonym for FEWER THAN.
LARGER	See OR LARGER.
NUMBER	The number of members of a set of atoms or bonds. Returns an integer (e.g. “MYVAR = THE NUMBER OF CHLORINE ATOMS ALPHA TO ATOM*1”).
OR LARGER	Used only in connection with ring size (e.g. “IF ATOM*1 IS IN A RING SIZE 7 OR LARGER ...”)
MORE THAN aaa	Tests the number of objects in a set (e.g. “IF THERE IS MORE THAN ONE CHLORINE ALPHA TO ATOM*1 ...”).
MORE*THAN aaa	Synonym for MORE THAN.
SIZE(S) n	Used only together with RING, to specify the ring size (e.g. “IF ATOM*1 IS IN A RING SIZE 5”; “IF ATOM*1 IS IN A RING OF SIZE 5 THROUGH 7”).
SIZE(S) n THROUGH n	

a) n represents a positive integer; aaa represents an English language word for a number between one and twenty (i.e. included in Table 12).

b) The values of properties are Boolean unless otherwise indicated.

c) BRIDGE and BRIDGED are synonymous. “BRIDGE(D)” (i.e. with “D” in parenthesis) is not part of the language.

Table 12. Named Numerical Values

Reserved word

ONE

TWO

THREE

FOUR

FIVE

SIX

SEVEN

EIGHT

NINE

TEN

ELEVEN

TWELVE

THIRTEEN

FOURTEEN

FIFTEEN

SIXTEEN

SEVENTEEN

EIGHTEEN

NINETEEN

TWENTY

Table 13. Logical Variables

Reserved word	Notes
FAIL	Boolean False (return value from a subroutine).
SUCCESS	Boolean True (return value from a subroutine).
SUCCESSFUL	Tests whether SUCCESS was returned by a preceding call to a subroutine.
UNSUCCESSFUL	Tests whether FAIL was returned by a preceding call to a subroutine.

Table 14. Operators^a

Reserved word ^b	Notes
-	Arithmetic subtraction.
%	Integer arithmetic modulus.
/	Arithmetic division.
+	Arithmetic addition.
=	Arithmetic equals symbol.
.EQ.	Relational equality operator.
.GE.	Relational greater than or equal operator.
.GT.	Relational greater than operator.
.LE.	Relational less than or equal operator.
.LT.	Relational less than operator.
.NE.	Relational not equal operator.
.OR.	Boolean operator for use in arithmetic conditional statements – e.g. “IF (VAR1 .EQ. 2 .OR. VAR1 = 3) THEN ...”.
ADD n	Increases the numerical rating for the transform by the specified amount. See the section entitled ‘Rating’.

AND:	See the section entitled ‘Conditional Statements’.
CLEAR	Empties an atom or bond set. Removes a previous designation (see DESIGNATE). E.g. “CLEAR SAVED*ATOM 1”; “CLEAR THE CURRENT*RING”)
CONTAINED	Tests whether a query is a member (or subset) of a set (e.g. “IF ATOM*3 IS CONTAINED IN THE CURRENT*RING ...”). See also SAME.
DECREMENT	Synonym for SUBTRACT (<i>q.v.</i>).
DESIGNATE	Assigns the contents of a set, or attaches a property to a set (e.g. “DESIGNATE AS THE CURRENT*RING THE HETEROCYCLIC RING CONTAINING ATOM*3”; “DESIGNATE THE GROUP ON SAVED*ATOM 2 AS INTERFERING”).
GETS	Initialises the contents of a set that has been declared by the writer (e.g. “MYSET GETS BENZYLIC ATOMS”).
INCREMENT	Synonym for ADD.
LOWER*RATING	See the section entitled ‘Rating’.
NO	Synonym of NOT (<i>q.v.</i>) for use in statements such as “IF THERE IS NO HYDROGEN ON ATOM*1...”.
NON	Tests whether the query object is a member of the set of objects not having a specified property (e.g. “IF ATOM*1 IS NON AROMATIC ...” means “if atom 1 is a member of the set of atoms that are not aromatic”). See also NOT.
NOT	Tests whether the query object is absent from a specified set of objects having a specified property (e.g. “IF ATOM*1 IS NOT AROMATIC ...” means “if atom 1 is not a member of the set of atoms that are aromatic”). See also NON.
OR:	See the section entitled ‘Conditional Statements’.
PERMUTE	Changes the formal bond order of a specified bond in an aromatic ring from single to double or <i>vice versa</i> , adjusting the orders of the other bonds in the aromatic system accordingly (e.g. “PERMUTE AT BOND*4”).
PUT:	Adds an object to a set (e.g. “PUT: THE ATOMS ALPHA TO ATOM*1 INTO MYATMS”).
RAISE*RATING	See the section entitled ‘Rating’.

REMOVE:	Removes a member from a set (e.g. “REMOVE: ATOM*1 FROM MYATMS”). Note the inclusion of the colon – see also REMOVE in Table 15.
SAME	In terms of program behaviour, SAME is a synonym of CONTAINED (i.e., it tests whether the query is a member of a specified set). However, to avoid confusing wording, SAME should be used only when the query is known to be a single object and the second reference is to a set that is known to have only one member (e.g. “IF ATOM*1 IS THE SAME AS SAVED*ATOM 4”, where it is known that there is only one atom in SAVED*ATOM 4; if that is not known, it is preferable to write “IF ATOM*1 IS CONTAINED IN SAVED*ATOM 4”).
SAVE AS n	The keying word is “SAVE” alone but it is normally followed by “AS” for clarity. 32 atom sets and 32 bond sets are available for use by the CHMTRN writer (e.g. “SAVE AS 5 THE ATOMS WITHIN GAMMA TO ATOM*2” would put those atoms into saved atom set 5).
SAVE*SEQUENTIALLY STARTING FROM n	Creates saved atom sets each containing a single member from a specified input set. The saved sets are numbered sequentially starting from n (e.g. “SAVE*SEQUENTIALLY STARTING FROM 3 THE ATOMS WITHIN GAMMA TO ATOM*2”).
SUBTRACT	Decreases the numerical rating for the transform by the specified amount. See the section entitled ‘Rating’.
THROUGH	Used only with SIZE(S), <i>q.v.</i>
TOGGLE BIT n	Changes BIT n in the SPECIAL*SET from 1 to 0 or <i>vice versa</i> . The value of n must be between 1 and 64
TOGGLE FLAG aaa	Changes the value of FLAG aaa from true to false or <i>vice versa</i> .
TURN OFF BIT n	Sets BIT n in the SPECIAL*SET to 0.
TURN OFF FLAG aaa	Sets the value of FLAG aaa to false.
TURN ON BIT n	Sets BIT n in the SPECIAL*SET to 1.
TURN ON FLAG aaa	Sets the value of FLAG aaa to true.
ZERO	The empty set (e.g. “MYSET GETS ZERO” makes MYSET an empty set).

- a) Operators used in mechanism statements are listed in Table 15.
- b) aaa represents a previously-declared name for a Boolean variable; n represents an integer.

Table 15. Terms used in mechanism statements

Reserved word	Notes
ANIONIZE	Assigns a negative charge to a specified atom.
ATTACH ^a	Attaches a functional group to a carbon atom, which becomes the origin of the group (e.g. ATTACH AN ALDEHYDE TO ATOM*3 makes atom 3 the carbon atom of a new aldehyde). See also INTRODUCE.
BREAK	Removes a bond of any order (e.g. BREAK BOND*1).
CATIONIZE	Assigns a positive charge to a specified atom.
DELETE	Removes an atom and the bonds connected to it (e.g. DELETE ATOM*1).
DEFINED*ANTI	Defines the stereo-relationship between substituents on a newly-formed double bond or the diastereomeric relationship between a pair of atoms and/or bonds one of which is on a new asymmetric centre. See the section entitled 'Stereochemistry'.
DEFINED*CIS	Defines the stereo-relationship between a newly-created stereocentre on a ring and an existing one (e.g. INTRODUCE A NITRILE ON ATOM*2 DEFINED*CIS TO ATOM*4; BOND*2 IS DEFINED*CIS TO BOND*4).
DEFINED*SYN	Defines the stereo-relationship between a newly-created stereocentre on a ring and an existing one (e.g. INTRODUCE A NITRILE ON ATOM*2 DEFINED*SYN TO ATOM*4; BOND*2 IS DEFINED*SYN TO BOND*4).
DEFINED*TRANS	Defines the stereo-relationship between substituents on a newly-formed double bond or the diastereomeric relationship between a pair of atoms and/or bonds one of which is on a new asymmetric centre. See the section entitled 'Stereochemistry'.

DOUBLE	Changes the order of a bond to double.
INTRODUCE ^a	Puts a functional group complete with its carbon atom origin, or a named substructure such as ISOPROPYL, onto an atom (e.g. INTRODUCE A KETONE ON ATOM*3 adds a complete ketone group, the carbon origin of which is thus alpha to atom 3). See also ATTACH.
INVERT	Inverts the tetrahedral stereochemistry at an atom in the precursor relative to that in the target (e.g. INVERT AT ATOM*3).
JOIN	1. Connects two atoms with a single bond. 2. If the atoms are already bonded, increases the bond order by one. The identifiers for the two atoms must be separated by “AND” (e.g. JOIN ATOM*1 AND ATOM*2).
NEUTRALIZE	Removes any charge from a specified atom.
RACEMIZE	Removes tetrahedral stereochemistry that was present at an atom in the target when forming the precursor (e.g. RACEMIZE AT ATOM*3).
RADICALIZE	Makes an atom into a radical.
REMOVE	Removes a functional group from a specified location (e.g. REMOVE THE KETONE ON ATOM*6). “GROUP” can be used in place of a specific group name, but this needs to be done with care to make sure there is no ambiguity, since there may be more than one group associated with a location. Note that REMOVE is not the same as REMOVE:, with a terminal colon (See Table 14).
RETAIN	Keeps the tetrahedral stereochemistry at an atom in the precursor the same as in the target (e.g. RETAIN AT ATOM*3).
SEPARATE	Removes completely any bonding between a pair of atoms (e.g. SEPARATE ATOM*1 AND ATOM*2).
SINGLE	Changes the order of a bond to single.
TRIPLE	Changes the order of a bond to triple.

a) If a group containing an alkyl group is attached or introduced, the alkyl group defaults to a carbon atom carrying no other carbon or heteroatoms. E.g. ATTACH AN ESTER TO ATOM*1 , causes ATOM*1 to be extended to C(=O)-O-C[HS=3].

Table 16. Control and conditional terms

Reserved word ^a	Notes
...	The contents of a line of code that follow are comments.
....	Marks the start and end of the mechanism statements (see the section entitled 'Mechanism Statements').
()	Used in the usual way in programming code that handles arithmetic and associated relational operations [e.g. "VAL = (V1 + V2 + V3)/ V4"; "IF (V1 .GT. V2 OR V3 .EQ. V4) THEN ..."]
AND:IF	See the section entitled 'Conditional Statements'.
AND*THEN	See the section entitled 'Conditional Statements'.
BEGIN aaa	See the section entitled 'Conditional Statements'.
BLKEND aaa	See the section entitled 'Conditional Statements'.
BRANCH	See the section entitled 'Control Statements'.
CALL aaa	See the section entitled 'Control Statements'.
DO aaa	See the section entitled 'Control Statements'.
END*TRANSFORM	See the section entitled 'Control Statements'.
ENDDO	See the section entitled 'Control Statements'.
EXIT aaa	See the section entitled 'Conditional Statements'.
FOR EACH	Causes an operation to be applied iteratively for members of a set of atoms or bonds. See the section entitled 'Control Statements'.
FOR*EACH	Synonym for FOR EACH.
FOREACH	Synonym for FOR EACH.
KILL	See the section entitled 'Control Statements'.
GO (TO) aaa	See the section entitled 'Control Statements'.
OR:IF	See the section entitled 'Conditional Statements'.
THEN	See the section entitled 'Conditional Statements'.

THEN*TO aaa	See the section entitled ‘Control Statements’.
TRANSFORM n	A unique integer identifier for the transform.
RETURN TRUE	Causes a subroutine to return control to the calling transform, returning SUCCESS.
RETURN FALSE	Causes a subroutine to return control to the calling transform, returning FAIL.
SUBROUTINE	The header for a block of code that is not part of a transform but can be called from any transform.
WARNING n:aaa	A message to be displayed to the end user. The integer, n, is a globally unique identifier for the message; aaa is the text of the message.

- a) aaa represents a string of characters; n represents a positive integer.

Table 17. Other reserved words

Reserved word	Notes
AND	Word separator required in statements such as “THE BOND SPANNING ATOM*2 AND ATOM*5”.
BIT n	One of the bits in the SPECIAL*SET (<i>q.v.</i>). n must be an integer between 1 and 64.
EMPTY	Used only in connection with the SPECIAL*SET (<i>q.v.</i>). “IF THE SPECIAL*SET IS EMPTY ...” returns true if all bits in the SPECIAL*SET are off (i.e. all bits are binary zero) and otherwise returns false.
FLAG aaa	A Boolean variable identified by the name aaa. Up to 64 flags could be set up in the original implementation of CHMTRN. There is no limit in the current implementation. Flags are global to the file in which they are declared. They can be declared explicitly at the head of the file, or declaration is implicit the first time the flag is referenced.
SPECIAL*SET	A 64-bit word available to the CHMTRN writer.
TOWARDS	Word separator required for use with statements about appendages. See the section entitled ‘Appendages’.

Table 18. Words Used in Declarative Statements

Reserved word	Notes ^a
SET(S) aaa, aaa ...	Declares a list of names of sets.
SET*ARRAY(S) aaa, aaa ...	Declares a list of names of arrays of sets.
VARIABLE(S) aaa, aaa ...	Declares a list of names of variables.
VARIABLE*ARRAY(S) aaa, aaa ...	Declares a list of names of arrays of variables.
a) Declaration of sets, arrays, and variables is optional in the current implementation.	

Table 19. Buzz Words

Reserved word

A

ALSO

AN

AND:FOR

ANOTHER

ANY

ANYWHERE

ARE

ARRAY

ARRAYS

AS

AT

ATOM

ATOMS

ATTACHED

AWAY
BE
BEARING
BETTER
BONDED
BY
CAN
CENTER
CENTRE
CHANGE
CHARACTER
CHARGE
CODE
CONTAIN
CONTAINING
CONTAIN
CONTAINS
COPY
DIMENSIONED
DOES
DUPLICATE
ELEMENT
ELEMENTS
EMPTY
EXACTLY
EXO
FACE

FOR
FROM
GROUP
GROUPS
HAS
HAVE
IDENTICAL
IF
IN
INCLUDING
INTO
ION
IS
LENGTH
LIST
MECHANISM
MEMBER
MEMBERS
MOLECULE
MOLECULES
MUST
NEXT
OF
OFFSPRING
OLD
OLEFIN*TYPE
ON

ONLY
ORDER
PARENT
PATH
PATHS
PRECURSOR
PRESENT
PREVIOUS
REQUIRED
SHORTEST
SIMULTANEOUSLY
STRUCTURAL
TARGET
TEST
THAT
THE
THERE
TO
UNIT
UNITS
UP
VALUE
VALUES
WHICH
WITH
WORD
WORDS

WRITE